

Asher Siddiqui

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EXPERIENCE

TAFSOL Technologies

Karachi, Pakistan (Hybrid)

Software Engineering Intern | Go, Echo, MySQL, Docker Compose, AWS

Mar. 2025 – Present

- Optimized login response through concurrent processing and query merging, resulting in a reduced response time from ~7s to ~1.5s and improved user experience.
- Developed a double debit prevention system using client provided transaction keys, eliminating duplicate charges and ensuring idempotent transactions.
- Implemented encryption using AES-256-CBC for securing wallet balances and historical wallet snapshots, ensuring confidentiality of wallet data at rest.
- Engineered an IP based rate limiting middleware using in-memory token bucket algorithm with endpoint specific custom quotas, preventing abuse on sensitive APIs with minimal latency.
- Built a real time chat system using WebSockets and events with status tracking and concurrent messaging, and reduced delivery time from ~1.3s to ~0.4s via LRU caching to eliminate database hits, resulting in an improved user experience.
- Designed and implemented a custom RBAC middleware with per-route permission enforcement, enabling flexible fine grained authorization across APIs.
- Developed device level authentication using device fingerprints, reducing account takeovers and enhancing user transparency through alert emails.
- Developed a Quick Pay feature using QR codes for internal network payments, reducing steps for transfers by 40%.
- Refactored utilities by replacing interface types with generics, improving type safety and reducing overall runtime panics.
- Implemented logging with file size based rotation, enabling compliance auditing and improving visibility.
- Integrated AWS S3 file storage using AWS SDK, enabling reliable file handling across the application.

Grayhat Private Limited

Islamabad, Pakistan (Remote)

Software Engineering Intern | JavaScript, Python

July 2024 – Sept. 2024

- Developed the core functionality for v0.dev's VS Code extension MVP, implementing AI generated UI component generation directly within the IDE.
- Developed an automated workflow tracker by integrating GitHub, Kimai, Activity Watch, and Plane, boosting developer productivity and visibility.

PROJECTS

FYP - DDoS Detection in Kubernetes using ML and DL | Kubernetes, Ansible, Prometheus, Falco, Go, Python | [GitHub](#)

- Deployed a bare metal Kubernetes cluster and a botnet using Ansible to create simulated DDoS attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline with Falco and Node Exporter to collect system calls and resource metrics, ensuring centralized logging and data generation.
- Developed and deployed attack scripts to simulate and orchestrate attacks via a C&C server and botnet nodes, enabling automated DDoS attacks.

Load Balancer | Go, Express.js, Docker, Git | [GitHub](#)

- Designed and implemented a high throughput HTTP reverse proxy implementing Round Robin and Least Connection algorithms using Go, achieving 2000 requests per second with minimal latency.
- Increased reliability by integrating health checks, ensuring automatic rerouting of traffic to available servers.
- Containerized the load balancer using Docker, enabling seamless deployment and scalability, and consistent performance across environments.

TECHNICAL SKILLS

Languages: Go, Python, Java

Backend & Frameworks: Echo, Spring Boot

Databases: MySQL, Oracle RDBMS, MongoDB, Redis

DevOps & Cloud: Docker, Docker Compose, Kubernetes, AWS, Linux, Bash, Prometheus, Node Exporter, Falco

EDUCATION

FAST - National University of Computer and Emerging Sciences (NUCES)
Bachelor of Science in Software Engineering - CGPA: 3.30/4.00
Relevant Courses: Applied DevOps, Cloud Security, Agile Software Project Management
Awards & Honors: Dean's List of Honors (3.5+ SGPA) in Fall 2023, Spring 2024, Fall 2024

Karachi, Pakistan
August 2021 – June 2025